

OPIPE-06: Cross-Scope Design Patterns

Series: OPIPE | Notebook: 6 of 6 | Created: March 2026 | Last Updated: 03/25/2026

Correlating Logs, Spans, Metrics, and Events Across Scopes

OpenPipeline's six scopes process data independently — but observability value comes from **correlating** across them. A slow API response (span) may be caused by a database timeout (log) that triggers a Davis problem (event) and degrades an SLO metric. This notebook covers design patterns that connect the dots across scopes, cascade processing for derived data, and a production readiness checklist.

For individual scope deep-dives, see **OPIPE-02** (spans), **OPIPE-03** (metrics), **OPIPE-05** (events). For log processing, see **OPLOGS-01**. For trace analysis, see **SPANS-04: Service Dependencies & Flow Analysis**.

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Prerequisites

Requirement	Details
Dynatrace Environment	SaaS with Grail
Permissions	<code>storage:logs:read</code> , <code>storage:spans:read</code> , <code>storage:metrics:read</code> , <code>storage:events:read</code> , <code>storage:bizevents:read</code>
Recommended	Complete OPIPE-01 through OPIPE-05

1. Shared Dimensions: The Correlation Key

Cross-scope correlation works when different data types share **common dimension values**. If your logs and spans both carry `service.name` and `k8s.namespace.name` , you can join them in DQL.

Correlation Fields Across Scopes

Field	Logs	Spans	Metrics	Events	Bizevents
<code>dt.entity.service</code>	Yes	Yes	Yes	Yes	—
<code>dt.entity.host</code>	Yes	—	Yes	Yes	—
<code>service.name</code>	Sometimes	Yes	—	—	—
<code>k8s.namespace.name</code>	Yes	Yes	Yes	—	—
<code>k8s.deployment.name</code>	Yes	Yes	Sometimes	—	—
<code>trace.id</code>	Sometimes	Yes	—	—	—
<code>dt.security_context</code>	If enriched	If enriched	If enriched	If enriched	If enriched

The Enrichment Strategy

If a critical dimension is missing from one scope, **add it via OpenPipeline enrichment** so that cross-scope queries work:

Scope	Missing Field	How to Add
Logs	service.name	Enrich from dt.entity.service via entity lookup
Spans	k8s.deployment.name	Enrich from k8s.pod.name by stripping replica suffix
Metrics	k8s.namespace.name	Enrich from entity tags or extension configuration

```
// Cross-scope: Correlate error logs with error spans by service entity
fetch logs, from:-1h
| filter loglevel == "ERROR" AND isNotNull(dt.entity.service)
| summarize error_logs = count(), by:{dt.entity.service}
| lookup [
  fetch spans, from:-1h
  | filter http.response.status_code >= 500 AND
isNotNull(dt.entity.service)
  | summarize error_spans = count(), by:{dt.entity.service}
], sourceField:dt.entity.service, lookupField:dt.entity.service, fields:
{error_spans}
| sort error_logs desc
| limit 10
```

2. Pattern: Same Metric from Multiple Scopes

A powerful validation pattern: extract the **same metric** from two different data sources and compare them. If they diverge, one pipeline has a configuration problem.

Example: Error Count from Logs vs. Spans

Source	Metric Key	Extraction Rule
Logs	log.error_count	Count of loglevel == "ERROR" , by dt.entity.service
Spans	span.error_count	Count of http.response.status_code >= 500 , by dt.entity.service

These metrics should track each other. If logs show 10x more errors than spans, it could mean:

- Application-level errors are logged but not reflected in HTTP status codes
- Span sampling is dropping error spans (see **OPIPE-03** on sampling-aware metrics)
- The log pipeline is catching errors from non-HTTP sources (background jobs, message consumers)

The comparison itself is diagnostic — the divergence tells you something about your data.

```
// Compare: Error counts from logs vs. spans by service
fetch logs, from:-1h
| filter loglevel == "ERROR" AND isNotNull(dt.entity.service)
| summarize log_errors = count(), by:{dt.entity.service}
| lookup [
    fetch spans, from:-1h
    | filter http.response.status_code >= 500 AND
isNotNull(dt.entity.service)
    | summarize span_errors = count(), by:{dt.entity.service}
], sourceField:dt.entity.service, lookupField:dt.entity.service, fields:
{span_errors}
| fieldsAdd ratio = if(isNotNull(span_errors) AND span_errors > 0,
    then: round(toDouble(log_errors) / toDouble(span_errors), decimals: 1),
    else: -1.0)
| sort log_errors desc
| limit 10
```

3. Pattern: Cascade Processing

Cascade processing creates a chain of derived data across scopes:

```
Span → generates Event → Event triggers Metric → Metric drives SLO
```

Example: Slow Transaction Cascade

Stage	Scope	Configuration	Output
1. Detect	Spans	Filter: <code>span.kind == "server" AND duration > 5s</code>	Matching spans
2. Generate	Spans → Events	Event generation rule: create event with <code>event.type = "slow_transaction"</code>	Event record per slow span
3. Extract	Events → Metrics	Metric extraction: <code>slow_transaction.count</code> by <code>service.name</code>	Metric time series
4. Alert	Metrics → SLO	SLO: <code>slow_transaction.count < 10</code> per 5m per service	SLO breach triggers alert

Design Considerations

- **Each stage adds latency** — The full cascade may take seconds from span ingestion to SLO evaluation
- **Failures propagate** — If the span pipeline drops the slow span, the event is never generated, and the metric never counts it
- **Test end-to-end** — After configuring a cascade, verify data flows through every stage

```
// Identify slow transactions that could trigger a cascade
fetch spans, from:-1h
| filter span.kind == "server" AND duration > 5000000000
| summarize slow_count = count(), avg_duration_ms = avg(toDouble(duration) / 1000000),
    by:{service.name}
| sort slow_count desc
| limit 10
```

4. Pattern: Unified Bucket Families

When related data across scopes should share the same lifecycle and access controls, use a **bucket family** naming convention:

Bucket Name	Scope	Contents	Retention
checkout_logs	Logs	Checkout service application logs	35 days
checkout_spans	Spans	Checkout service trace data	14 days
checkout_events	Events	Checkout-related Davis events	35 days
checkout_bizevents	Bizevents	Purchase and cart business events	90 days

Benefits

- **Consistent naming** — `checkout_*` makes it obvious which buckets belong together
- **Unified access control** — IAM policies can use `dt.system.bucket` patterns with wildcards
- **Coordinated retention** — Related data expires in a logical sequence (spans first, then logs, business events last)

Querying Across a Bucket Family

```
// Query all buckets in a family to understand data distribution
fetch logs, from:-1h
| filter matchesValue(dt.system.bucket, "checkout_*")
| summarize log_count = count()
| append [
  fetch spans, from:-1h
  | filter matchesValue(dt.system.bucket, "checkout_*")
  | summarize span_count = count()
]
```

5. When NOT to Use OpenPipeline

OpenPipeline is powerful but not always the right tool. Prefer DQL query-time processing when:

Scenario	Why Not OpenPipeline	Better Approach
Exploratory analysis	You do not know what patterns you are looking for yet	DQL ad-hoc queries
Frequently changing logic	Pipeline changes require deployment; DQL changes are instant	DQL with dashboard variables
Complex joins	OpenPipeline cannot join across scopes at ingestion	DQL <code>lookup</code> and <code>join</code> at query time
One-time investigations	Configuring a pipeline for a single investigation is overhead	DQL <code>parse</code> with DPL patterns
Historical data	OpenPipeline only processes new data — it cannot reprocess historical records	DQL for retroactive analysis
Correlation dashboards	Cross-scope correlation requires data from multiple tables	DQL with <code>append</code> and <code>lookup</code>

The Rule of Thumb

OpenPipeline for permanent, irreversible actions (drop, mask, route, extract). **DQL for everything else.**

If you are unsure whether a transformation belongs in OpenPipeline or DQL, ask: *"Will I regret not having the raw data?"* If yes, keep it raw and process at query time.

6. Production Readiness Checklist

Before deploying a new or modified OpenPipeline configuration to production:

Pipeline Design

- [] Each pipeline has a single, clear purpose (one source type per pipeline)
- [] Routing rules are ordered from most specific to least specific
- [] Default pipeline catches only unmatched data with short retention
- [] No overlapping routing conditions between pipelines

Processing Rules

- [] Masking rules applied BEFORE any other processing
- [] Drop rules filter noise before extraction (cost efficiency)
- [] Parsing rules tested against representative samples
- [] Enrichment fields have defined values for all conditions (no nulls)

Metric Extraction

- [] Cardinality estimated for each extracted metric (see **OPIPE-04**)
- [] No unique-per-record dimensions (`trace.id` , `span.id` , `content`)
- [] Sampling-aware extraction for span-derived count metrics (see **OPIPE-03**)
- [] Metric keys follow naming convention: `scope.metric_name` (e.g., `span.request_count`)

Security & Governance

- [] `dt.security_context` set on all scopes that need access control
- [] Security events routed to dedicated long-retention buckets
- [] Compliance-relevant data never dropped or sampled
- [] PII masking verified on all scopes that handle user data

Testing

- [] End-to-end data flow verified: ingestion → pipeline → bucket → query
- [] Cascade chains tested: span → event → metric → SLO (if applicable)
- [] Volume monitoring in place to detect unexpected spikes or drops
- [] Rollback plan documented (revert to previous pipeline configuration)

Summary

In this notebook you learned:

- **Shared dimensions** — Cross-scope correlation requires common fields like `dt.entity.service` and `k8s.namespace.name`

- **Same metric from multiple scopes** — Extract matching metrics from logs and spans to validate pipeline health
 - **Cascade processing** — Chain span → event → metric → SLO for automated detection and alerting
 - **Unified bucket families** — Name related buckets with a common prefix for consistent lifecycle management
 - **When NOT to use OpenPipeline** — Keep data raw for exploratory analysis, changing logic, and one-time investigations
 - **Production readiness** — Checklist covering pipeline design, processing rules, metric extraction, security, and testing
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What's Next?

You have completed the OPIPE series. From here:

- **OPLOGS** — Deep-dive into log processing: **OPLOGS-01: OpenPipeline Fundamentals**
 - **SPANS** — Master trace analysis: **SPANS-01: Fundamentals**
 - **OPMIG** — Migrating from classic logs: **OPMIG-01: Why Migrate**
 - **ORGNZ** — Data organization and governance: **ORGNZ-01: Introduction**
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References

- [OpenPipeline Documentation](#)
 - [Grail Data Lakehouse](#)
 - [DQL Cross-Data Queries](#)
 - [SLO Configuration](#)
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This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.